

Physically Unclonable Functions: A Survey and Our Direction

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Abstract. PUF is an important hardware security device that can be used for authentication and encryption key generation. PUFs, also known as device fingerprints, can generate random digital values that can be regenerated using physical quantities that are correlated with manufacturing variations. PUF can be used to prevent the distribution of counterfeit products, to authenticate IoT devices at low cost, and to realize more secure communication technologies. In the academic field, various PUFs using various physical quantities that depend on manufacturing variations have been proposed. This paper briefly surveys PUFs that have been proposed in the academic field after describing the basics of PUF. After that, we show our contribution and research direction.

1. Introduction

Nowadays, the distribution of counterfeit goods is expanding in all fields. It is said that the global market for counterfeit goods reaches 10 billion dollars, and it has been reported that not only financial damage and damage to brand image, but also national defense and life are in danger in some cases [1]. The semiconductor industry is no exception, and the problem of semiconductor counterfeit products has become a major problem in recent years. They say that the globalization of the semiconductor design and manufacturing industry, and changes in the procurement methods of semiconductor users are the background to the distribution of semiconductor counterfeit products [2,3]. The distribution of semiconductor counterfeit products causes leakage of personal information and software information in chips, and hacking of devices.

Assignment of unique authentication ID to each device is a conventional countermeasure against semiconductor counterfeits. A unique authentication ID for device authentication is written to the non-volatile memory such as EEPROM and flash memory of each device, and a genuine or imitation product is identified by this information. However, this method has the disadvantage of being vulnerable to side-channel attacks and easy to replicate. On the other hand, the cost of using tamper-resistant memory that is resistant to side-channel attacks is high.

PUF is a circuit technology that complements the drawbacks of conventional authentication technologies and performs individual authentication using variations in the manufacturing of devices that are physically difficult to replicate. Variations in device manufacturing results in variations in the delay time of electrical signals, leakage currents, the initial state of the memory, and so on. PUFs use these physical quantities to generate device-specific bit strings. This bit string is also called a device fingerprint because it is reproducible and reflects the manufacturing variability inherent in the device.

This paper shows a brief survey of PUF and our research direction. The rest of the paper is organized as follows. Section 2 describes the basics of PUF. Section 3 briefly surveys PUFs proposed in academic research area. After that, in section 4, we show our contribution and research direction. Finally, section 5 concludes the paper.

2. Basics

2.1 System-Level Model of PUF

Fig. 1 shows the system-level model of PUF. PUF can be modeled at the system level as a challenge-response system that is a black box with a challenge input c and a response output r [4]. Conceptually expressed in mathematical terms, output r can be regarded as a function of f with input c . Unlike normal function, output r depends not only on c but also on device-specific manufacturing variations.

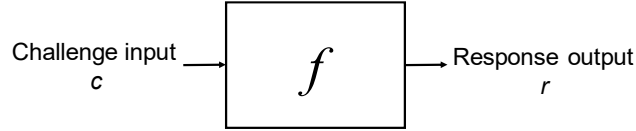


Fig. 1. System-level model of PUF

2.2 Generation of Unique Response Output

A challenge input common to the PUFs on each device is input to the PUFs to obtain response outputs. Since the output response of the PUF depends on the challenge input and the variation in device manufacturing, the response output is expected to be unique to each device. Let the i th challenge input be c_i and the 1-bit response output of c_i be r_i . The response output $r_1 r_2 \dots r_n$ of the challenge input $c_1 c_2 \dots c_n$ is expected to be a device-specific n -bit bit string. In Fig. 2, a common challenge input $c_1 c_2 \dots c_n$ is input to PUFs on Device 1, Device 2, ..., Device m to generate n -bit unique response outputs $r_{11} r_{12} \dots r_{1n}$, $r_{21} r_{22} \dots r_{2n}$, ..., $r_{m1} r_{m2} \dots r_{mn}$.

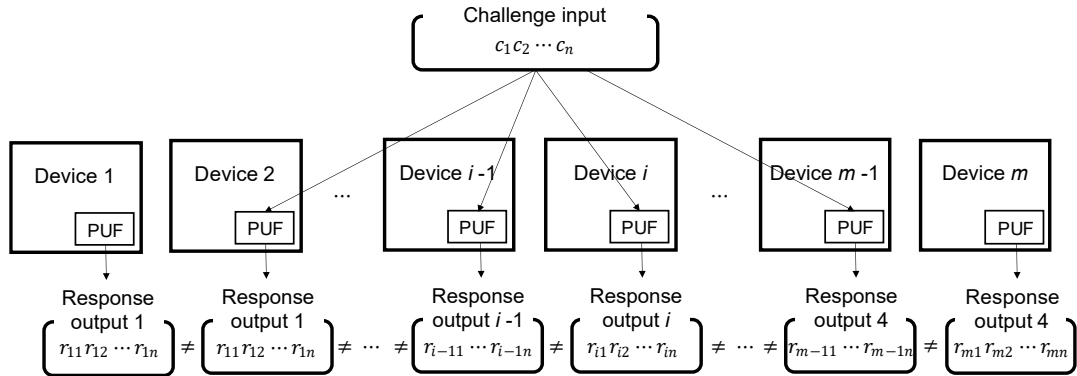


Fig. 2. Generation of unique n -bit response outputs with common challenge input

2.3 Strong PUF and Weak PUF

PUFs are classified into strong PUFs and weak PUFs based on the relationship between the number of challenge input and response output pairs (CRPs) and the chip area. Strong PUF is a PUF in which the number of CRPs is on the order of an exponential with respect to the chip area, and a weak PUF is a PUF in which the CRPs are of a linear order with respect to the chip area [5]. The main application of strong PUF is device authentication. The main application of weak PUF is IP protection and the generation of private keys used for encrypted data communication using public key authentication [4,6].

Both PUF require enough uniqueness. Since weak PUFs are used for cryptographic application, weak PUFs must have higher reliability than strong PUFs. Since strong PUFs are used for device authentication of numerous products, strong PUFs must generate longer-length response output. As a result, strong PUFs must support huge number of CRPs with small chip area [7].

3. Classification of PUF

This section explains the typical PUFs that have been proposed so far.

3.1 Delay-Based PUF

Delay-based PUF is a PUF that uses the delay variation of the gate circuit. Typical delay-based PUFs are Arbiter PUF and Ring Oscillator (RO) PUF [7]. Arbiter PUF is a PUF that uses the variation of the delay difference of two identical delay lines due to process variation. Fig. 3 (a) shows a typical Arbiter PUF.

RO PUF is a PUF that uses the variation of the oscillation frequency of the ring oscillator due to the process variation. Fig. 3 (b) shows a typical RO PUF.

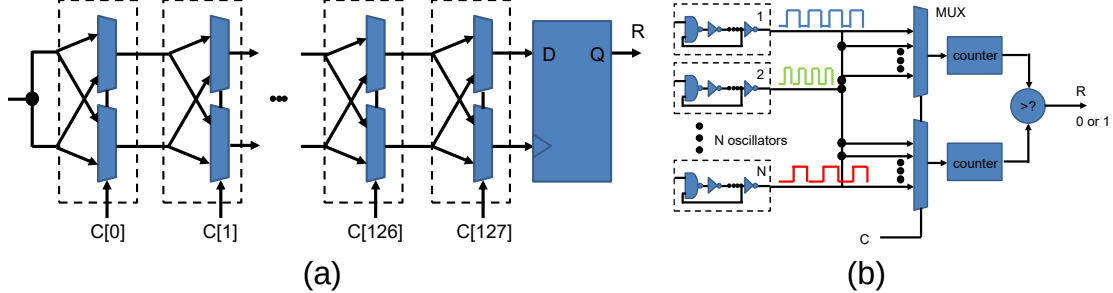


Fig. 3. Typical delay-based PUFs: (a) Arbiter PUF (b) RO PUF

Arbiter PUF is a strong PUF, while RO PUF is a weak PUF [4]. In addition to them, a strong PUF using a reconfigurable RO called Loop PUF has been proposed [8]. Delay-based PUFs have the advantage of being easy to implement because they are simple circuits composed of standard cells, but they have the disadvantage of being vulnerable to aging degradation. The effects of aging on delay-based PUFs have also been studied [9].

3.2 Memory-Based PUF

Memory-based PUFs are PUFs that use variations in the characteristics of memory cells in memory. The most famous one is SRAM PUF [10,11]. SRAM PUF is based on the fact that the initial value of the SRAM cell, which stores the value in a cross-coupled loop, depends on the variation (Fig. 4 (a)) [12,13]. DRAM PUFs use variations in the capacitance of memory cells. Variations in memory cell capacitance surface in various ways. As a result, various DRAM PUFs have been proposed [14-17]. Retention-based DRAM PUF is one of the most famous ones [18,19]. This PUF utilizes the fact that decay of charge of DRAM cells over time after charging is unique in each device (Fig. 4 (b)). PUFs using emerging non-volatile memory has also been proposed [21,22]. These PUFs take advantage of the variation in the resistance value of the memory cell. In particular, STT MRAM PUF has the advantages of high security, high power efficiency, and high bit density [22]. On the other hand, it also has the disadvantage of low reliability. In general, memory-based PUFs are used as weak PUF.

3.3 Other PUFs

Various other PUFs have been proposed. Chowdhury et al. proposed a weak PUF using the characteristic variation of the null convention logic gate, which is a logic gate in an asynchronous circuit [23]. The inverter/amplifier analog PUF, which is a weak PUF, uses the variation of threshold voltage and switching voltage of inverter and inverting amplifier [5,24]. Xi et al. proposed a strong PUF that uses the variation of the subthreshold current of a CMOS device as an entropy source [25]. Chatterjee et al. proposed a RF-PUF using variations in analog and RF characteristics due to process variations in wireless transmitters (Tx) [26]. The ADC PUFs use variations of the capacitor array inside

CDAC [27-29]. The RF-PUF and ADC PUFs are assumed to be for device authentication. Therefore, the reliability of them can be lower than that of weak PUFs.

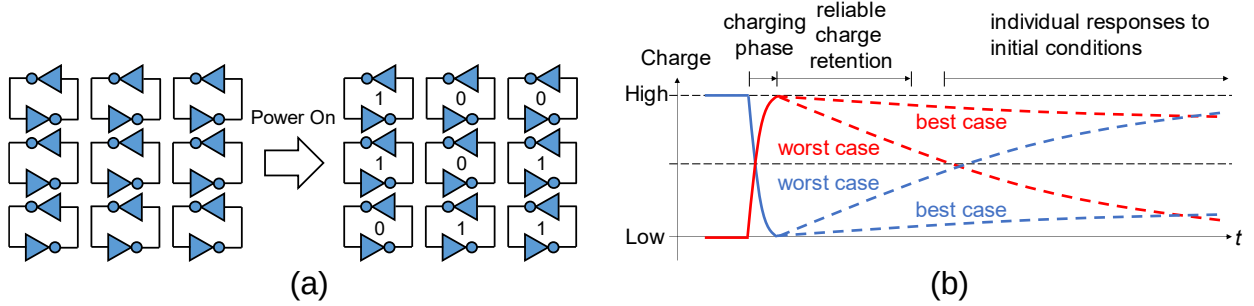


Fig. 4. (a) Basic idea of SRAM PUF (b) Charge decay of DRAM cell over time

3.4 FPGA PUF

FPGAs can reconfigure digital circuits. Consequently, PUFs that can be implemented with digital devices can be configured on FPGAs for FPGA device authentication and cryptographic key generation (FPGA PUFs). PUFs configured on FPGAs include RO PUF, Arbiter PUF, and Loop PUF [7,8,30,31]. In addition, in FPGAs, latching circuits can be used to construct a cross-coupled loop like a SRAM cell. PUFs can be realized by configuring the multiple cross-coupled loops in an array, which is called Butterfly PUF [6]. Whether a FPGA PUF is strong PUF or weak PUF depends on the implemented PUF.

4. Our Contribution and Direction

We have been studying the use of a time-to-digital converter (TDC) as a PUF (TDC PUF). Like ADC PUF, the variation of some of the components of the circuit is used as the PUF. In the case of flash TDC PUF, the buffer delay of each stage varies. The variation of the buffer delay of flash TDC is used as the PUF.

As a first step, we proposed a method using flash TDC with a histogram calibration function as a PUF [32]. This TDC can accurately estimate the resolution variation of each stage using the calibration function. This variation is unique to each circuit. Output a response output from the obtained delay variation and challenge input. Fig. 5 shows our four-stage proposed TDC PUF. As far as we know, this is the first proposal for TDC PUF.

Like ADC PUFs, the method of using an existing circuit as a PUF operates as a normal circuit in normal operating mode. Therefore, even if extra circuit is added for PUF, the effective area overhead can be kept lower than the conventional method. TDC PUFs using buffer delay variations are classified as delay-based PUFs. TDC can measure delays with high resolution during normal operation. By using this function, it is possible to realize a PUF with higher performance than the conventional delay-based PUF. Our group will continue the research in this direction.

5. Conclusion

In this paper, we briefly surveyed the major PUFs proposed in the academic field. We also described our contribution and research direction. We will continue our research based on the direction we have proposed.

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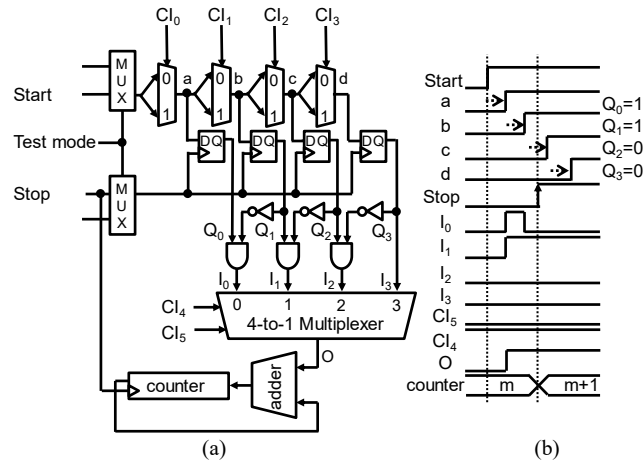


Fig. 5. TDC PUF using TDC with linearity self-calibration [32] @ IEEE Copyright

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